Markowitz, Black-Litterman and Risk Parity on Ten US Stocks

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1 The Core Idea

The repository builds three long-only portfolios from the same ten US stocks and compares them. The first is a Markowitz mean-variance portfolio, shown at maximum Sharpe ratio and at minimum variance. The second is a Black-Litterman portfolio that combines market equilibrium returns with two investor views. The third is a risk parity portfolio in which every stock contributes the same share of total risk. All three use daily prices from 2019 to 2026 and a risk-free rate of 3%.

- **Same data, different weights:** the maximum Sharpe portfolio puts 70% in two stocks (NVDA and JNJ), while risk parity keeps every stock between 6.1% and 14.7%.
- **Returns drive the answer:** the expected returns are historical averages, so the Sharpe ratios are in-sample and optimistic.

2 Why It Matters

- **The optimiser amplifies estimation error.** The annual mean of NVDA is 66.9% with a standard error of about 18 points. If its mean were 50% instead, its maximum Sharpe weight would fall from 39.2% to 22%, and at 30% it would be zero.
- **Weights are not risk.** In the maximum Sharpe portfolio NVDA is 39.2% of the money and 72.8% of the variance. In risk parity every stock is about 10% of the variance and none is more than 14.7% of the money.
- **An in-sample Sharpe ratio is a ceiling.** Fitted on 2019–2022 and tested on 2022–2026, the maximum Sharpe portfolio goes from 1.11 to 1.02. Fitted on the second half and tested on the first, it goes from 2.30 to 0.67, below equal weights at 0.80.
- **The prior matters as much as the views.** The Black-Litterman weights come mostly from the equilibrium returns, not from the two views. The portfolio's own expected return under the posterior is 11.5% (8.5% above the risk-free rate), against the 23.7% in the results table, which uses historical means.

3 Assumptions

Every item below is built into `portfolio_optimizer.py`. The last sentence of each says what moves if it is wrong.

- **Data.** Yahoo Finance adjusted closes for 10 stocks from 2 January 2019 to 31 July 2026, with rows that have any missing close dropped: 1,905 prices, 1,904 returns, 7.6 years. Adjusted closes include dividends, which is right for total return. A different window changes every mean, and the second half of this one was much stronger than the first.
- **Returns.** Daily simple returns, annualised with 252 days: $\mu = 252\,\mathbb{E}[r]$ and $\Sigma = 252\,\mathrm{Cov}(r)$. This treats returns as independent with constant moments. Volatility clustering and fat tails make the true risk larger in stress.
- **Expected returns.** The historical mean, with no forecast and no shrinkage. The standard error of an annual mean is the volatility divided by $\sqrt{7.6}$: 7 to 18 points. Differences of a few points between stocks, such as JNJ at 13.8% and PG at 10.8%, are inside the noise.
- **Covariance.** The sample covariance, with no shrinkage and fixed over the whole period. The average pairwise correlation is 0.365, and correlations tend to rise when markets fall, so the diversification shown is a fair-weather figure.
- **Constraints.** Long-only, fully invested, each weight at most 100%, a constant risk-free rate of 3%, no borrowing, no costs, no taxes and no rebalancing. Costs and taxes would lower every figure, most for the portfolios that trade most.
- **Solver.** SLSQP from equal weights. I checked that the solver reports success for the maximum Sharpe, minimum variance, risk parity, Black-Litterman and 40 frontier problems, and that 20 random starting points (Dirichlet draws, seed 0) reach weights within 0.3 percentage points for maximum Sharpe and Black-Litterman and within 0.08 for risk parity. Minimum variance, whose objective is flat in the low-volatility stocks, stays within 1.2 points. The solutions do not depend on the starting point beyond these differences.
- **Black-Litterman inputs.** Equal market weights stand in for capitalisation, risk aversion is 2.5 and $\tau = 0.05$. The two views (NVDA beats the average of AAPL and MSFT by 5%, JPM returns 8%) are chosen for the example. With another prior, such as market-cap weights, the equilibrium returns of the large technology names would change.
- **Return basis.** The equilibrium returns Π and the two views are excess returns, so the posterior is an excess return. The code adds the 3% risk-free rate back before the maximum Sharpe step, which subtracts it once, so the Black-Litterman inputs sit on the same total-return basis as the historical means used for Markowitz. A view such as "JPM returns 8%" is therefore read as 8% above the risk-free rate, 11% in total.
- **Reporting.** The table of results computes the expected return, volatility and Sharpe of every portfolio with the historical means and covariance, including the Black-Litterman portfolio, whose weights come from other means. The return shown for that portfolio (23.7%) is not its model expectation (11.5% in total).
- **Frontier.** Forty target returns spread evenly between the lowest and highest single-stock mean (10.8% to 66.9%), so the frontier ends at 100% NVDA.

4 Method

4.1 Data

Daily adjusted closes come from Yahoo Finance through yfinance, for AAPL, MSFT, JPM, XOM, JNJ, PG, V, NVDA, KO and UNH. The request runs from 1 January 2019 to 1 August

2026, an end date that yfinance treats as exclusive. Columns are put in the ticker order straight after the download, so weights, covariance matrix and Black-Litterman views refer to the same asset order. With r the vector of daily simple returns, the annualized inputs use 252 trading days:

$$\mu = 252\,\mathbb{E}[r], \qquad \Sigma = 252\,\mathrm{Cov}(r).$$

4.2 Markowitz

For weights w, the expected return is $w^\top\mu$, the volatility is $\sqrt{w^\top\Sigma w}$ and the Sharpe ratio is $(w^\top\mu - r_f)/\sqrt{w^\top\Sigma w}$, with $r_f = 3\%$. The maximum Sharpe portfolio solves

$$\max_w \frac{w^\top\mu - r_f}{\sqrt{w^\top\Sigma w}} \quad \text{subject to} \quad \sum_i w_i = 1, \quad 0 \le w_i \le 1,$$

and the minimum variance portfolio minimizes $w^\top\Sigma w$ under the same constraints. Both use SLSQP from equal weights. The efficient frontier is traced by minimizing variance for 40 target returns spaced evenly between the lowest and highest stock mean. The maximum Sharpe weight of a stock rises with its mean and falls with its variance and its correlation with the other holdings. The minimum variance portfolio ignores μ and holds the low-volatility stocks.

4.3 Black-Litterman

The equilibrium excess return is $\Pi = \lambda\,\Sigma w_m$, with risk aversion $\lambda = 2.5$. The market weights w_m are equal, 10% per stock, as a stand-in for market capitalization. The views sit in a 2×10 matrix P and a vector Q. The first view puts $+1$ on NVDA and -0.5 on each of AAPL and MSFT, with $Q_1 = 5\%$. The second puts $+1$ on JPM, with $Q_2 = 8\%$. The view uncertainty is $\Omega = \mathrm{diag}(P\,\tau\Sigma\,P^\top)$ with $\tau = 0.05$, and the posterior mean is

$$\mu_{BL} = \left[(\tau\Sigma)^{-1} + P^\top\Omega^{-1}P\right]^{-1}\left[(\tau\Sigma)^{-1}\Pi + P^\top\Omega^{-1}Q\right].$$

This posterior is an excess return, like Π and Q. The code adds the risk-free rate back, $\mu_{BL} + r_f$, so that the maximum Sharpe step (which subtracts r_f once) sees it on the same footing as the historical means in the Markowitz problem. The maximum Sharpe portfolio is then solved with $\mu_{BL} + r_f$ and the same Σ. With no views $\mu_{BL} = \Pi = \lambda\Sigma w_m$, whose tangency portfolio is proportional to w_m, so the result is exactly the equal market weights. Because Ω is proportional to τ, the factor τ cancels in μ_{BL}: I ran $\tau = 0.01$, 0.05, 0.25 and 1 and the weights are identical. The formula gives each view the same confidence as the prior has in the same combination of stocks. Only Q, P and λ move the result.

4.4 Risk parity

Stock i contributes $RC_i = w_i(\Sigma w)_i/(w^\top\Sigma w)$ to portfolio variance, and the contributions sum to one. The weights minimize $\sum_i (RC_i - 1/10)^2$ subject to the budget constraint $\sum_i w_i = 1$ and bounds $0.001 \le w_i \le 1$, again with SLSQP. For uncorrelated stocks the answer would be proportional to $1/\sigma_i$. Here the correlations move it, and the high-volatility stocks (NVDA at 50.7%) get the smallest weights.

5 Method Comparison

Method	Strength	Weakness
Max Sharpe	Best risk-adjusted return in the sample	Concentrated; extremely sensitive to μ
Min variance	Needs no return forecast	Concentrated in defensive stocks; ignores return
Black-Litterman	Anchored on equilibrium; views are explicit	Result depends on prior and view scale
Risk parity	Diversified by risk; no return forecast	Ignores returns; leans on low-volatility names

When the return forecast is weak, which is the usual case, the methods that do not use one (minimum variance, risk parity) are the safer default, and equal weights are a fair benchmark. Black-Litterman is useful when the investor has a view with a stated size, and it needs a prior that people accept. Maximum Sharpe is a diagnostic of what the data would say if taken at face value.

6 Results

Portfolio	Expected return	Volatility	Sharpe
Max Sharpe (Markowitz)	37.4%	25.1%	1.37
Min variance (Markowitz)	15.6%	15.8%	0.79
Black-Litterman	23.7%	18.8%	1.10
Risk parity	20.9%	17.5%	1.02
Equal weights (comparison)	23.8%	18.9%	1.10

The first four rows are the repository output. The equal-weight row is recomputed with the repository functions. The maximum Sharpe portfolio has the highest ratio, as it must, because it maximises that number on the data it is measured on. Risk parity (1.02) is below equal weights (1.10) in this sample: it moves money toward JNJ, PG and KO, which had low volatility and also low returns. Black-Litterman (1.10) matches equal weights because its weights are close to equal. Minimum variance has the lowest return and the lowest volatility, and five stocks hold 96% of it.

Stock	Max Sharpe	Min variance	Black-Litterman	Risk parity
AAPL	10.4	0.0	10.6	8.0
MSFT	0.0	10.3	10.6	8.6
JPM	2.5	0.3	6.0	8.5
XOM	10.3	9.7	10.5	10.0
JNJ	31.2	33.8	10.4	14.7
PG	0.0	21.9	10.5	13.4
V	0.0	0.0	10.4	8.7
NVDA	39.2	1.3	10.1	6.1
KO	6.4	20.5	10.4	12.9
UNH	0.0	2.2	10.4	9.1

Weights in percent, rounded to one decimal from the unrounded solver output.

The maximum Sharpe portfolio holds six stocks, so its weights are sensitive to the estimated means. Black-Litterman stays within about 0.6 points of the 10% equal prior in nine stocks and

gives JPM 6.0%. Risk parity is the most spread out. Recomputed from the saved weights, its ten risk contributions lie between 9.97% and 10.03%, so the solver meets its target.

The Black-Litterman result needs a careful reading. Without views the optimiser returns the equal market weights it started from, 10% in each stock, because $\Pi = \lambda \Sigma w_m$ is the return vector that makes w_m the tangency portfolio. The equilibrium returns already imply that NVDA beats the average of AAPL and MSFT by 5.4%, and that JPM earns 9.8%. The two views (5% and 8%) are below those values, by 0.4 and 1.8 points, so they act as a small tilt: NVDA moves from 10.0% to 10.1%, which is almost nothing, while JPM falls from 10.0% to 6.0% because 8% is well below its implied 9.8%. The other eight weights lie between 10.4% and 10.6%. NVDA has the highest implied return (16.0%) because the equilibrium return rises with covariance and NVDA has the highest variance, but the same variance penalises it in the optimiser, so its weight stays at the prior. A real bullish view would move it: with Q_1 at 10%, 15% and 20% the NVDA weight is 17.1%, 24.2% and 31.5%.

Per-stock inputs behind these weights (percent per year, recomputed from the same download):

	AAPL	MSFT	JPM	XOM	JNJ	PG	V	NVDA	KO	UNH
Mean	32.7	25.4	23.9	19.9	13.8	10.8	17.4	66.9	13.3	14.3
Volatility	30.9	29.3	29.5	31.0	19.3	20.2	25.8	50.7	19.9	33.1
Std. error of mean	11.2	10.7	10.7	11.3	7.0	7.4	9.4	18.4	7.3	12.0
Share of variance, max Sharpe	8.9	0.0	1.5	5.1	9.8	0.0	0.0	72.8	1.9	0.0

The means are measured with errors of 7 to 18 points, which is as large as most of the gaps between stocks. The portfolio that is built on them inherits that noise.

Holdout test. The 1,904 returns split into 952 each (2019-01-03 to 2022-10-12 and 2022-10-13 to 2026-07-31). Each portfolio is fitted on one half and measured on the other. Sharpe ratios, with the historical means of the measured half for every portfolio (recomputed with the repository functions):

	Fit 2019–22, test 2022–26		Fit 2022–26, test 2019–22	
Portfolio	In sample	Out of sample	In sample	Out of sample
Max Sharpe	1.11	1.02	2.30	0.67
Min variance	0.52	0.91	1.53	0.66
Black-Litterman	0.87	1.56	1.84	0.74
Risk parity	0.76	1.64	1.64	0.75
Equal weights	0.80	1.76	1.76	0.80

Equal weights beat every optimised portfolio out of sample in both directions, but the gap to risk parity, the closest, is small: 1.76 against 1.64, and 0.80 against 0.75. The fitted weights also move a great deal between halves: maximum Sharpe is 70% AAPL in the first half and 22% NVDA, 26% KO, 25% JNJ and 22% JPM in the second. The test says that the weights estimated on one period are not stable, not that equal weights are better in general: the second half was a strong market for all of them (equal-weight Sharpe 1.76).

7 Worked Example

The maximum Sharpe weights are the four-decimal values in the repository output. The annual means come from the same price download. Each stock adds $w_i \mu_i$ to the expected return, and

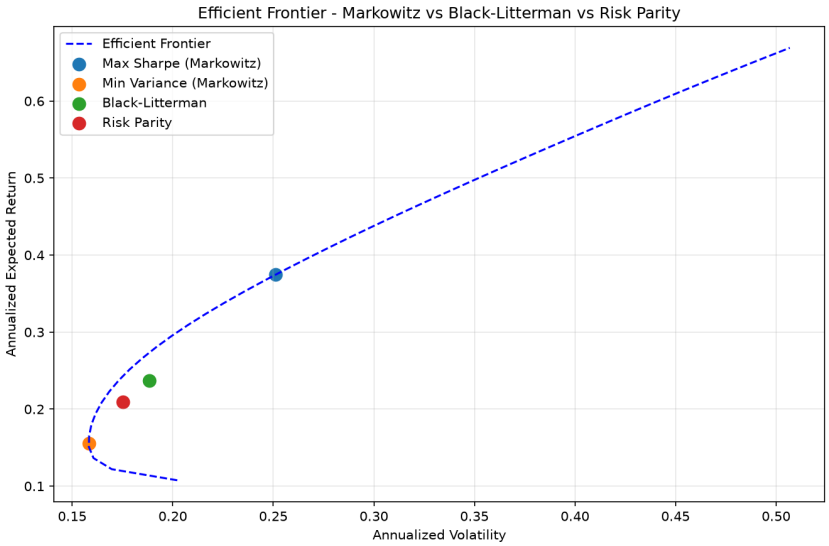


Figure 1: Efficient frontier (dashed) and the four portfolios, from the repository output.

only six weights are nonzero:

$$\begin{aligned}\mu_p &= 0.3922 \times 0.6691 + 0.3123 \times 0.1384 + 0.1035 \times 0.3269 + 0.1029 \times 0.1991 \\ &\quad + 0.0644 \times 0.1328 + 0.0247 \times 0.2387 \\ &= 0.2624 + 0.0432 + 0.0338 + 0.0205 + 0.0086 + 0.0059 = 0.3744.\end{aligned}$$

NVDA alone adds 0.2624, which is 26.2 points of the 37.4% expected return. With the stored volatility of 25.1%, the Sharpe ratio is

$$\frac{0.3744 - 0.03}{0.2510} = 1.372 \approx 1.37.$$

Both figures match `portfolio_stats.csv`. The risk side of NVDA: the portfolio variance is $0.2510^2 = 0.0630$, the covariance of NVDA with the portfolio is $(\Sigma w)_{NVDA} = 0.1169$, and so

$$RC_{NVDA} = \frac{0.3922 \times 0.1169}{0.0630} = 72.8\%.$$

Sensitivity. Keep everything else and change NVDA's mean μ_{NVDA} in the optimisation. The maximum Sharpe portfolio then holds:

μ_{NVDA} (%)	30	40	50	60	66.9	80
NVDA weight (%)	0.0	9.9	22.0	32.4	39.2	50.2
Sharpe	1.06	1.08	1.15	1.27	1.37	1.59

Each 10 points of NVDA return moves its weight by about 8 to 12 points (8.4 on the last step, from 66.9 to 80). One standard error of the mean (18 points) takes the position from 39% to 20%. Removing NVDA from the universe gives a Sharpe ratio of 1.06, with AAPL at 41.8% and JNJ at 21.2%. The 1.37 headline is therefore one stock's past: without that stock, the best portfolio is close to the 1.10 of equal weights that include it.

8 Limits and Extensions

- **In-sample means:** the expected returns and the risk come from the same history. NVDA's mean of 66.9% a year supplies most of the maximum Sharpe return, so the

result depends on one stock's past.

- **Illustrative views:** the two Black-Litterman views are chosen for the example, not forecasts. The NVDA view is close to what the prior already implies, so NVDA hardly moves (10.0% to 10.1%), while the JPM view is well below the implied 9.8% and cuts that weight from 10.0% to 6.0%.
- **Market weights:** equal weights replace market capitalization in the equilibrium return.
- **View basis:** the views are read as excess returns, like the equilibrium return, and the risk-free rate is added back afterwards. A forecaster who means a total return of 8% for JPM would have to enter 5%.
- **Covariance and solver:** Black-Litterman changes only the mean vector, and risk always uses the historical covariance. SLSQP reports success on every problem here, but the code does not check the flag.
- **Scope:** the portfolios are long-only, with no transaction costs and no rebalancing. These are model outputs, not recommendations.

Three extensions would fix the largest limits. First, shrink the means toward a common value, or put a cap of 20% to 25% on each weight, so that one noisy estimate cannot set the portfolio. Second, a rolling window with a rebalancing date, so that the holdout test above becomes the main result and turnover is measured. Third, market-capitalisation weights and an explicit view confidence in place of the proportional Ω, so that the strength of a view is a choice and not a by-product of the formula.

9 Conclusion

The four portfolios differ much less in what they know than in what they ask of their inputs. Maximum Sharpe needs the means to be right to within a few points, which the data cannot give, while risk parity and minimum variance need only the covariance. A portfolio that must survive its own estimation error should lean on the second group, and use the first as a sensitivity test.